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Aayush Mishra

Baltimore, USA | amishr24@jhu.edu | +1 667-379-9669 | aamixsh.github.io | aamixsh

`Towards efficient and reliable adaptation in AI models.`

Education

Johns Hopkins University — `Jan 2022 -`
- Ph.D. in Computer Science — `- ongoing`
- M.S.E. in Computer Science – GPA: 4.0/4.0 (transcript) — `- Dec 2023`

Indian Institute of Technology Mandi — `Aug 2015 - Jun 2019`
- B.Tech. in Computer Science and Engineering – GPA: 8.88/10.0 (transcript)
  with Minor in Management

Selected Publications

- Steered LLM Activations are Non-Surjective — 2026
  Aayush Mishra, Daniel Khashabi, Anqi Liu [pre-print, (Sci4DL, Re-Align)@ICLR]
- Genomic Next-Token Predictors are In-Context Learners — 2026
  Nathan Breslow, **Aayush Mishra**, et al. [TMLR, (Sci4DL, MLGenX (Oral))@ICLR]
- `IA2`: Alignment with ICL Activations Improves Supervised Fine-Tuning — 2025
  Aayush Mishra, Daniel Khashabi, Anqi Liu [ICLR]
- ICL CIPHERS: Quantifying "Learning" in ICL via Substitution Ciphers — 2025
  Zhouxiang Fang, **Aayush Mishra**, et al. [EMNLP (Main)]
- ODD: Overlap-aware Estimation of Model Performance under Distribution Shift — 2025
  Aayush Mishra, Anqi Liu [UAI]
- EigenLoRA: Recycle trained Adapters for Resource Efficient Adaptation and Inference — 2024
  Aayush Mishra*, Prakhar Kaushik*, et al. [pre-print]
- Do pretrained Transformers Learn In-Context by Gradient Descent? — 2024
  Aayush Mishra*, Lingfeng Shen*, Daniel Khashabi [ICML (Oral)]
- Source-Free and Image-Only Unsupervised Domain Adaptation for Category Level OPE — 2024
  Prakhar Kaushik, **Aayush Mishra**, et al. [ICLR]
- VStegNET: Video Steganography Network using Spatio-Temporal features and Micro-Bottleneck — 2019
  Aayush Mishra*, Suraj Kumar*, et al. [BMVC]

Industry Experience

Adobe — `May 2024 - Aug 2024`
Research Intern [Document Intelligence Lab]
- Developed EigenLoRA, to recycle pre-trained LoRAs for efficient inference and training of new LoRAs.

Microsoft
Data & Applied Scientist [Bing Shopping] — `Aug 2021 - Jan 2022`
- Developed 1) a query → product class (∼18k classes) model; 2) a relevance metric to improve offer ranking.

Siemens
Research Engineer [Automation] — `Jul 2019 - Aug 2021`
- Developed a CNN explainability and compression algorithm aimed at edge deployment.
- Used GAN-Dissection to procedurally generate traffic images for stress-testing traffic monitors (YOLO).
- Trained an RL agent to find edge-case safety violations of self-driving agents in a simulator.
- Developed a semantic search tool for codebases using Latent Semantic Analysis, and a Siemens social network post classification tool using statistical NLP. Work done during internships during 2017–2018.

Skills

Python	●●●●●
C++	●●●●○
Data Engineering	●●●●●
ML toolkits (`torch/TRL/etc.`)	●●●●●